

DEVILS ISLAND, WI, USA

WMO: 994190

Lat: 47.0795N

Lon: 90.7283W

Elev: 629

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-1.5	-13.3	2.7	-4.0	-9.1	3.5	0.7	39.9	19.3	36.5	16.2	19.6	270	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	78.7	69.0	75.8	67.2	73.3	65.9	71.5	75.4	69.5	73.2	67.8	71.7	9.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	113.9	73.8	68.4	106.7	72.1	66.3	99.3	70.0	35.8	75.5	34.1	73.2	32.5	71.8	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
39.9	34.2	29.9	-10.7	86.9	8.2	3.3	-16.6	89.3	-21.4	91.2	-25.9	93.1	-31.9	95.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	17.9	18.8	28.9	38.2	46.5	55.4	64.6	65.9	58.9	47.5	34.9	25.1
	DBStd	18.26	11.13	10.61	9.96	6.37	6.38	6.35	6.34	4.90	6.68	6.96	8.84	8.67
	HDD50	4415	996	874	656	357	150	21	0	7	129	453	772	
	HDD65	8564	1461	1294	1118	804	575	293	85	48	203	544	902	1237
	CDD50	1505	0	0	3	3	40	184	454	494	274	52	1	0
	CDD65	178	0	0	0	0	5	74	77	20	2	0	0	0
	CDH74	471	0	0	0	0	5	29	227	148	59	3	0	0
	CDH80	63	0	0	0	0	1	6	33	16	7	0	0	0
	WSAvg	14.0	16.3	16.8	14.9	15.0	12.9	10.6	10.5	11.3	12.2	15.3	16.0	16.2
	PrecAvg	28.9	1.3	0.9	1.3	2.2	2.9	3.7	3.5	3.0	3.3	3.5	2.2	1.6
Precipitation	PrecMax	37.6	2.6	2.2	3.1	8.1	5.0	8.1	8.1	6.0	7.2	8.6	4.4	3.5
	PrecMin	21.9	0.4	0.2	0.4	0.5	0.8	0.5	1.5	1.0	0.9	1.1	0.7	0.2
	PrecStd	4.2	0.6	0.5	0.7	1.6	1.1	1.7	1.7	1.2	1.5	1.9	1.1	0.8
	PrecStd	4.2	0.6	0.5	0.7	1.6	1.1	1.7	1.7	1.2	1.5	1.9	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.7	44.4	61.7	60.9	72.7	78.2	84.3	82.4	80.4	71.7	55.4	42.9	
	0.4% MCWB	38.6	37.9	55.4	49.3	59.4	67.1	69.6	71.1	68.9	62.0	49.3	40.0	
	2% DB	37.8	37.1	49.5	54.4	66.2	72.5	79.2	77.7	74.8	65.5	50.9	39.5	
	2% MCWB	35.6	32.7	44.8	44.7	55.5	61.9	69.5	69.6	67.1	57.3	45.3	37.6	
	5% DB	35.0	34.1	43.1	50.3	60.8	68.7	76.3	75.0	70.9	60.4	48.4	37.6	
	5% MCWB	33.4	31.5	38.3	43.6	51.7	60.8	67.4	67.5	65.0	54.2	44.2	36.0	
	10% DB	32.4	32.1	39.4	46.1	56.6	65.5	73.5	72.9	68.2	56.3	46.2	36.1	
	10% MCWB	30.5	30.1	36.0	40.8	49.3	59.4	66.1	66.4	63.0	51.4	42.8	34.6	
	0.4% WB	41.2	37.8	57.4	52.4	64.2	69.2	74.8	74.4	71.3	63.6	51.7	41.3	
	0.4% MCDB	42.7	41.5	62.4	59.2	68.7	74.7	79.6	78.5	75.6	70.3	52.6	42.5	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.6	33.5	44.3	47.2	57.9	65.1	71.7	71.7	68.5	58.0	47.3	38.4	
	2% MCDB	38.5	35.1	48.3	52.9	63.7	68.1	75.7	75.2	72.8	63.1	49.5	39.3	
	5% WB	34.1	31.9	39.0	44.0	54.1	62.1	69.4	69.8	66.1	54.5	45.0	36.6	
	5% MCDB	35.6	33.5	41.5	49.1	60.6	66.2	73.0	73.3	69.7	58.2	47.5	37.5	
	10% WB	31.6	30.3	36.7	41.5	51.1	59.9	67.5	68.1	63.7	52.0	42.9	34.9	
	10% MCDB	33.4	32.0	38.9	45.8	56.4	64.0	71.8	71.6	67.0	55.1	45.6	36.0	
Mean Daily Temperature Range	MDBR	9.6	10.0	9.4	9.5	12.1	13.1	11.2	8.9	8.5	7.3	7.7	7.9	
	5% DB	10.3	10.4	13.7	18.1	21.0	19.3	15.2	12.3	13.4	14.1	9.6	8.4	
	5% WB	9.7	8.3	9.5	11.0	12.0	11.4	8.6	7.1	7.9	9.3	8.1	7.7	
	MCDBR	10.9	9.2	12.5	16.6	19.9	17.5	13.3	11.2	12.1	12.6	8.8	7.3	
	MCWBR	10.2	8.1	9.8	11.1	14.2	12.9	9.2	7.9	8.5	9.1	8.2	7.2	
	MCWBR	10.2	8.1	9.8	11.1	14.2	12.9	9.2	7.9	8.5	9.1	8.2	7.2	
Clear-Sky Solar Irradiance	taub	0.260	0.279	0.306	0.346	0.371	0.380	0.387	0.381	0.364	0.318	0.295	0.272	
	taud	2.506	2.426	2.401	2.373	2.403	2.418	2.440	2.466	2.482	2.567	2.540	2.515	
	Ebn at Noon	272	286	292	288	284	280	277	274	269	268	254	252	
	Edh at Noon	20	26	31	36	36	36	35	32	29	23	19	17	
All-Sky Solar Radiation	RadAvg	406	720	1109	1457	1704	1838	1933	1624	1211	733	412	300	
	RadStd	44	73	101	151	122	127	78	107	73	84	52	31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page